

CPSC 304 Project Cover Page

Milestone #: 1

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Group Number: 75

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above. (In the case of Project Milestone 0, the main purpose of this page is for you to let us know your e-mail address, and then let us assign you to a TA for your project supervisor.)

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

2. A brief project description:

- a. The domain of our application is **animal shelter management**. This involves various aspects of caring for and managing animals in shelters, including tracking animal information, recording the adoption process, overseeing shelter operations, and handling donations, veterinary services, and volunteer/employee information.
- b. The database for our animal shelter system is designed to model the following aspects of the domain:
 - **Shelter branches:** Stores information about each branch's location, including donors, supplies, veterinarians, cages, and animals. Branch is an owner entity to the cage entity, as each cage can only have
 - **Animal Profiles:** Detailed profiles for each animal, including their ID, name, age, breed, and admission date. If a cat is brought in, its profile might also include specific details about its napping habits, while a dog's profile could indicate its guarding behavior.
 - **Adoption Process:** Captures adoption details such as adoption dates, fees and adopter's profile(ID, name, age, criminal records, address)
 - **Shelter Operations:** Maintains information about cages, including animal assignments, locations, and sizes.
 - **Donor Information:** Records details about donors, types of donations, and amounts.
 - **Veterinary Services:** Documents information about veterinarians and their hourly rate.
 - **Volunteer Information:** Tracks volunteers' profiles, their roles, and hours.
 - **Supplies Management:** Maintains inventory of supplies across branches, including supply names and associated quantities.
 - **Employee Information** section stores staff details, and salaries.

For instance, in a real-life situation, if a new animal is brought into the shelter, the system can quickly update its profile, assign it to an appropriate cage, and record provided veterinary services. Additionally, when donors and volunteers help out at the shelter, we create a record maintaining a profile history. Hence the database addresses critical needs of the animal shelter by creating records and maintaining a history of each activity ensuring smooth operations, efficient resource management, and effective care for the animals.

3. The functionality of the database:

The database for the animal shelter provides a range of functionalities that enhance the management and operation of the shelter across its various branches.

- Users can track detailed information about different types of animals with each animal having a unique ID, name, age, breed, cage information, admission date and health treatment information.
- When animals are adopted, users can also find information about adopters and animals adopted by them as well as other information regarding the adoption process.

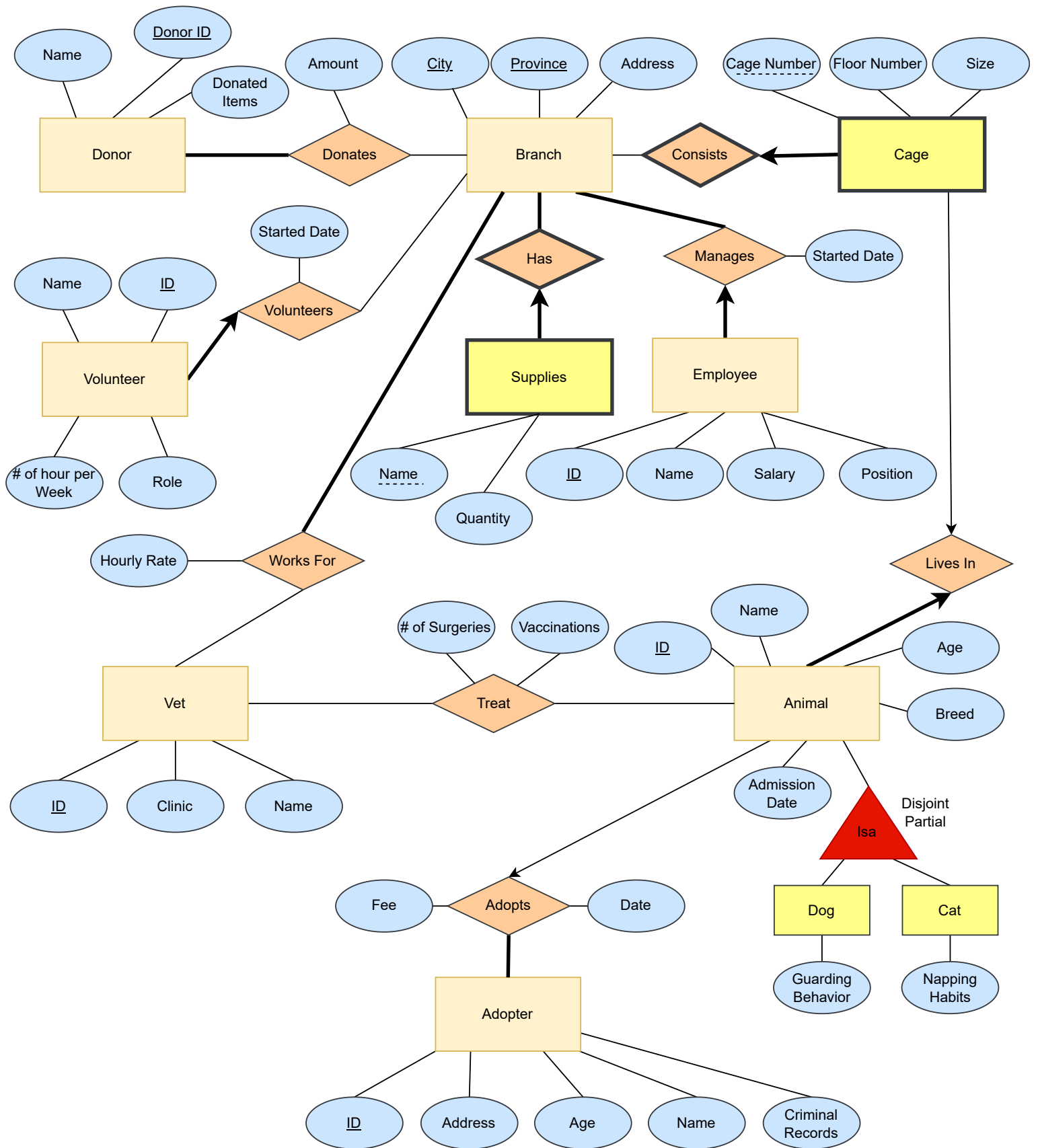
- Users can also record and access the information about different branches such as donors, volunteers, vets and employees that are associated with each branch
- Additionally, the database keeps an inventory of supplies at each branch, including pet food stock, toys, medical equipment, and cleaning supplies.

Overall, this comprehensive functionality supports efficient shelter operations and enhances the overall experience for animals, adopters, and staff alike.

4. Description of the application platform:

- a. In our project, we will use the department-provided **Oracle** for database management.
- b. For the technology stack, we are planning to use **JavaScript** as our primary programming language.

5. ER Diagram



7. Explanation for participation constraints:

- A Branch may consist of multiple cages, while a Cage must be assigned to only one Branch
- A Branch must manage some employees, while each employee is stationed at only one Branch
- A Branch must have some supplies inventory while Supplies must be assigned to some Branch
- Branches may have multiple Donors, and a Donor must donate to some Branch.
- Branches may have multiple Volunteers, and each Volunteer must volunteer at only one Branch.
- A Vet may work for multiple Branches, however, a Branch must have some Vet working in it.
- A Vet may treat multiple animals, and an animal may be treated by multiple Vets
- An Adopter must adopt some animal(s), while an animal may be adopted by only one adopter.
- An Animal must live in only one Cage, and a Cage may contain at most one animal.

ISA Structure Explanation:

Our animal shelter management system consists of two primary animal types - Cats and Dogs. However, this does not mean that only Cats and Dogs are present in the animal shelter, as it may include other types of animals in lesser frequencies, such as Birds. Hence, we have a **PARTIAL DISJOINT ISA** Structure.

Weak Entities Explanation:

A Cage relies on its unique Cage Number to be identified within each Branch, and relies on the Branch City and Province, along with its cage number to be uniquely identified. Hence, Cage is a weak entity with Branch as its owner entity

Supplies rely on the Branch entity to be uniquely identified with primary key consisting of City, Province, Supplies Name. Hence Supplies is a weak entity with Branch as the owner entity.